Department of Computer Science

Faculty of Computing

Project Task 5

(IP Addressing)

Programme	:	Bachelor of Computer Science (*Data Engineering*)
Subject Code	:	SECR1213
Subject Name	:	Network Communications
Session-Sem	:	2024/2025-1

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Section	:	12	
Group	:	12.3 – 4Fabs	
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5.1 Network address given

Group	Network Address
12.3	192.164.0.0/8

The given network address of our group is 192.164.0.0/8. It provides over 16 million IPs (2^24) which making it suitable for large-scale networks with thousands of devices. This ensures future scalability for massive expansion.

Network Address details:

IP address (decimal)	192.164.0.0
IP address (binary)	11000000 10100100 00000000 00000000
Subnet mask (decimal)	255.0.0.0
Subnet Mask (binary)	11111111 00000000 00000000 00000000

We divided the subnet mask into the network portion (8 bits) and host portion (24 bits). By using the AND operation of the IP address with the subnet mask, we identified the following:

Network Address	192.0.0.0
Broadcast Address	192.255.255.255
Range of Usable Addresses	192.0.0.1 – 192.255.255.254

5.2 Divide it in the best possible way for network.

By using the Classless Inter-Domain Routing (CIDR) method, we decided to divide the network into separate subnets for each work area outlined in task 4, except for the server room and staff office which share the same subnet. A total of 8 subnets are needed to accommodate the design of our network. Thus, we need to borrow 3 bits from the host portion of the IP address, as 2^3 = 8 subnets are necessary to meet the network's segmentation requirements.

By borrowing these 3 bits, the subnet mask changed form /8 to /11.

Network Portion = Original 8 bits + 3 borrowed subnet bits = 11 bits

Host Portion = Total 32 bits – 11 Network Portion bits = 21 bits

This approach ensures that each subnet and devicecs and device has sufficient IP addres ses while maintaining future scalability and efficient resourse allocation across the faculty's network infrastucture.

Divide subnet to 8 subnets:

Subnet Network	Network portion	Subnet bits	Host portion		IP address
0	1100 0000	000	0 0000 0000 0000 0000 0000	Network Address	192.0.0.0
			1 1111 1111 1111 1111 1111	Broadcast Address	192.31.255.255
1	1100 0000	001	0 0000 0000 0000 0000 0000	Network Address	192.32.0.0
			1 1111 1111 1111 1111 1111	Broadcast Address	192.63.255.255
2	1100 0000	010	0 0000 0000 0000 0000 0000	Network Address	192.64.0.0
			1 1111 1111 1111 1111 1111	Broadcast Address	192.95.255.255
3	1100 0000	011	0 0000 0000 0000 0000 0000	Network Address	192.96.0.0
			1 1111 1111 1111 1111 1111	Broadcast Address	192.127.255.255

4	1100 0000	100	0 0000 0000 0000 0000 0000	Network Address	192.128.0.0
			1 1111 1111 1111 1111 1111	Broadcast Address	192.159.255.255
5	1100 0000	101	0 0000 0000 0000 0000 0000	Network Address	192.160.0.0
			1 1111 1111 1111 1111 1111	Broadcast Address	192.191.255.255
6	1100 0000	110	0 0000 0000 0000 0000 0000	Network Address	192.192.0.0
			1 1111 1111 1111 1111 1111	Broadcast Address	192.223.255.255
7	1100 0000	111	0 0000 0000 0000 0000 0000	Network Address	192.224.0.0
			1 1111 1111 1111 1111 1111	Broadcast Address	192.255.255.255

Distribution of subnet for different work area:

Subnet	Work Area	Network Address	Broadcast Address	Range of usable address	Custom subnet mask
0	Server Room and Staff Office	192.0.0.0	192.31.255.255	192.0.0.1 - 192.31.255.254	/11 255.224.0.0
Ground Floor					
1	General Lab 1	192.32.0.0	192.63.255.255	192.32.0.1 - 192.63.255.254	/11 255.224.0.0
2	General Lab 2	192.64.0.0	192.95.255.255	192.64.0.1 - 192.95.255.254	/11 255.224.0.0
3	Cisco Networking Lab	192.96.0.0	192.127.255.255	192.96.0.1 - 192.127.255.254	/11 255.224.0.0
4	Embedded Lab	192.128.0.0	192.159.255.255	192.128.0.1 - 192.159.255.254	/11 255.224.0.0
First Floor					

5	Hybrid Classroom	192.160.0.0	192.191.255.255	192.160.0.1 - 192.191.255.254	/11 255.224.0.0
6	Student Lounge	192.192.0.0	192.223.255.255	192.192.0.1 - 192.223.255.254	/11 255.224.0.0
7	Video Conferencing Room	192.224.0.0	192.255.255.255	192.224.0.1 - 192.255.255.254	/11 255.224.0.0

Network distribution for workstation in different work areas.

Ground Floor:

1. General Lab 1

Device	IP Address
PC1	192.32.0.1
PC2	192.32.0.2
PC3	192.32.0.3
PC4	192.32.0.4
PC5	192.32.0.5
PC6	192.32.0.6
PC7	192.32.0.7
PC8	192.32.0.8
PC9	192.32.0.9
PC10	192.32.0.10
PC11	192.32.0.11
PC12	192.32.0.12
PC13	192.32.0.13
PC14	192.32.0.14
PC15	192.32.0.15
PC16	192.32.0.16
PC17	192.32.0.17
PC18	192.32.0.18
PC19	192.32.0.19
PC20	192.32.0.20
PC21	192.32.0.21
PC22	192.32.0.22
PC23	192.32.0.23
PC24	192.32.0.24
PC25	192.32.0.25
PC26	192.32.0.26
PC27	192.32.0.27
PC28	192.32.0.28
PC29	192.32.0.29

PC30	192.32.0.30
PC31	192.32.0.31
Projector 1	192.32.0.32
WAP 1	192.32.0.33
WAP 2	192.32.0.34

2. General Lab 2

Device	IP Address
PC32	192.64.0.1
PC33	192.64.0.2
PC34	192.64.0.3
PC35	192.64.0.4
PC36	192.64.0.5
PC37	192.64.0.6
PC38	192.64.0.7
PC36	192.64.0.8
PC40	192.64.0.9
PC41	192.64.0.10
PC42	192.64.0.11
PC43	192.64.0.12
PC44	192.64.0.13
PC45	192.64.0.14
PC46	192.64.0.15
PC47	192.64.0.16
PC48	192.64.0.17
PC49	192.64.0.18
PC50	192.64.0.19
PC51	192.64.0.20
PC52	192.64.0.21
PC53	192.64.0.22
PC54	192.64.0.23
PC55	192.64.0.24
PC56	192.64.0.25
PC57	192.64.0.26
PC58	192.64.0.27
PC59	192.64.0.28
PC60	192.64.0.29
PC61	192.64.0.30
PC62	192.64.0.31
Projector 2	192.64.0.32
WAP 3	192.64.0.33

3. Cisco Networking Lab

Device	IP Address
PC63	192.96.0.1
PC64	192.96.0.2
PC65	192.96.0.3
PC66	192.96.0.4
PC67	192.96.0.5
PC68	192.96.0.6
PC69	192.96.0.7
PC70	192.96.0.8
PC71	192.96.0.9
PC72	192.96.0.10
PC73	192.96.0.11
PC74	192.96.0.12
PC75	192.96.0.13
PC76	192.96.0.14
PC77	192.96.0.15
PC78	192.96.0.16
PC79	192.96.0.17
PC80	192.96.0.18
PC81	192.96.0.19
PC82	192.96.0.20
PC83	192.96.0.21
PC84	192.96.0.22
PC85	192.96.0.23
PC86	192.96.0.24
PC87	192.96.0.25
PC88	192.96.0.26
PC89	192.96.0.27
PC90	192.96.0.28
PC91	192.96.0.29
PC92	192.96.0.30
Projector 3	192.96.0.31
WAP 4	192.96.0.32

4. Embedded Lab

Device	IP Address
PC93	192.128.0.1
PC94	192.128.0.2
PC95	192.128.0.3
PC96	192.128.0.4
PC97	192.128.0.5
PC98	192.128.0.6

PC99	192.128.0.7
PC100	192.128.0.8
PC101	192.128.0.9
PC102	192.128.0.10
PC103	192.128.0.11
PC104	192.128.0.12
PC105	192.128.0.13
PC106	192.128.0.14
PC107	192.128.0.15
PC108	192.128.0.16
PC109	192.128.0.17
PC110	192.128.0.18
PC111	192.128.0.19
PC112	192.128.0.20
PC113	192.128.0.21
PC114	192.128.0.22
PC115	192.128.0.23
PC116	192.128.0.24
PC117	192.128.0.25
PC118	192.128.0.26
PC119	192.128.0.27
PC120	192.128.0.28
PC121	192.128.0.29
PC122	192.128.0.30
PC123	192.128.0.31
Projector 4	192.128.0.32
WAP 5	192.128.0.33

Ground floor with label:

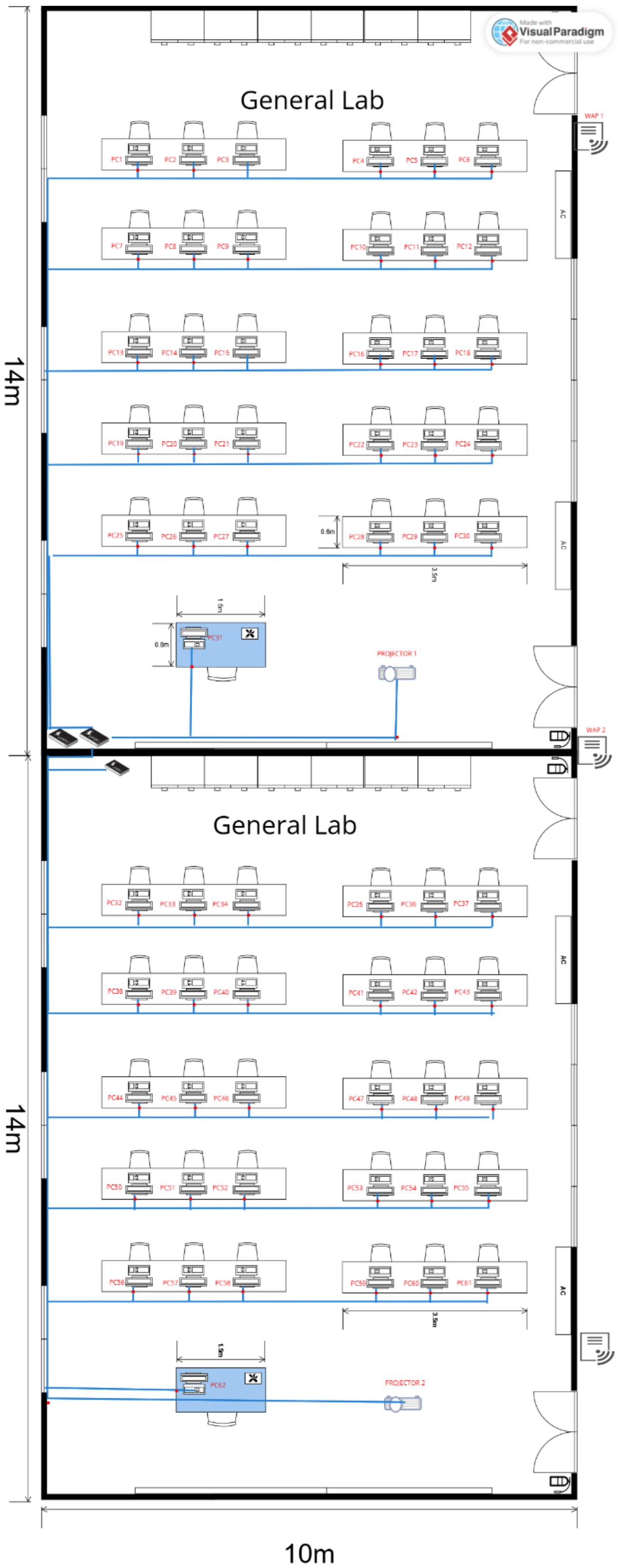

Figure 5.2.1 General Lab 1 & 2 with label

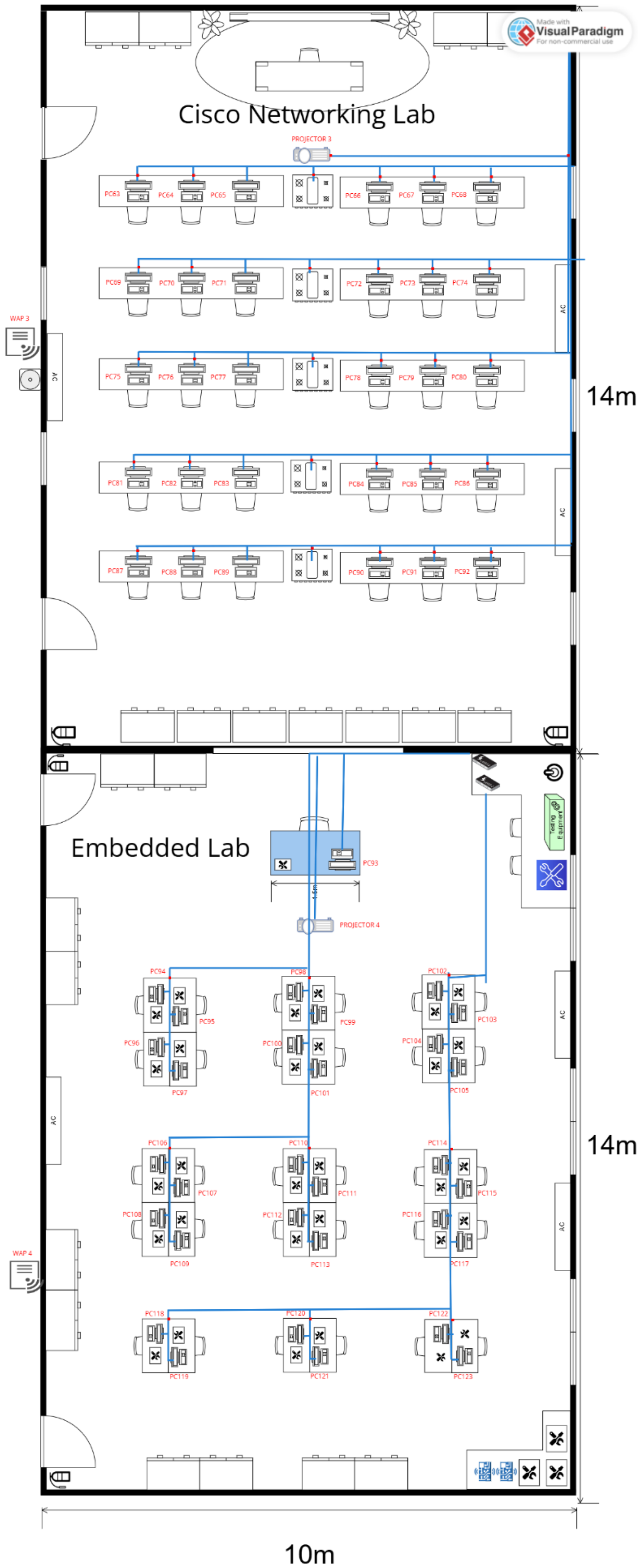

Figure 5.2.2 Cisco Networking Lab and Embeded Lab with label

First Floor:

5. Hybrid Classroom

Device	IP Address
PC124	192.160.0.1
PC125	192.160.0.2
PC126	192.160.0.3
PC127	192.160.0.4
PC128	192.160.0.5
PC129	192.160.0.6
PC130	192.160.0.7
PC131	192.160.0.8
PC132	192.160.0.9
PC133	192.160.0.10
PC134	192.160.0.11
PC135	192.160.0.12
PC136	192.160.0.13
PC137	192.160.0.14
PC138	192.160.0.15
PC139	192.160.0.16
PC140	192.160.0.17
PC141	192.160.0.18
PC142	192.160.0.19
PC143	192.160.0.20
PC144	192.160.0.21
PC145	192.160.0.22
PC146	192.160.0.23
PC147	192.160.0.24
PC148	192.160.0.25
PC149	192.160.0.26
PC150	192.160.0.27
PC151	192.160.0.28
PC152	192.160.0.29
PC153	192.160.0.30
PC154	192.160.0.31
Projector 5	192.160.0.32
WAP 6	192.160.0.33

6. Student Lounge

Device	IP Address
WAP 7	192.192.0.1

7. Video Conferencing Room

Device	IP Address
IoT 1	192.224.0.1
WAP 8	192.224.0.2

8. Staff Office

Device	IP Address
PC155	192.0.0.1
PC156	192.0.0.2
PC157	192.0.0.3
PC158	192.0.0.4
PC159	192.0.0.5
PC160	192.0.0.6
PC161	192.0.0.7
PC162	192.0.0.8
PC163	192.0.0.9
PC164	192.0.0.10
WAP 9	192.0.0.11

9. Server Room

Device	IP Address
Router 1	192.0.0.12
Router 2	192.0.0.13
Router 3	192.0.0.14
Router 4	192.0.0.15
Server 1	192.0.0.16
Server 2	192.0.0.17
Server 3	192.0.0.18
Server 4	192.0.0.19
WAP 10	192.0.0.20

First floor with label:

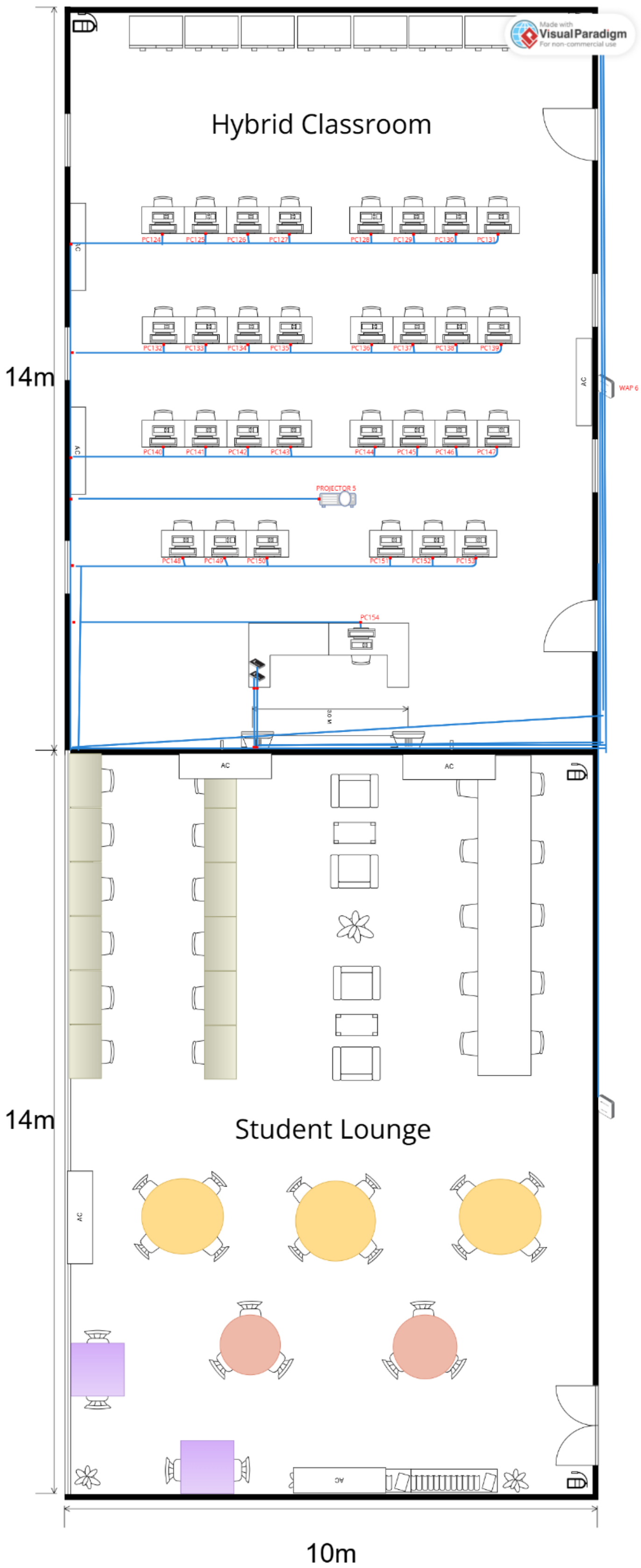

Figure 5.2.3 Hybrid Classroom and Student Lounge with label

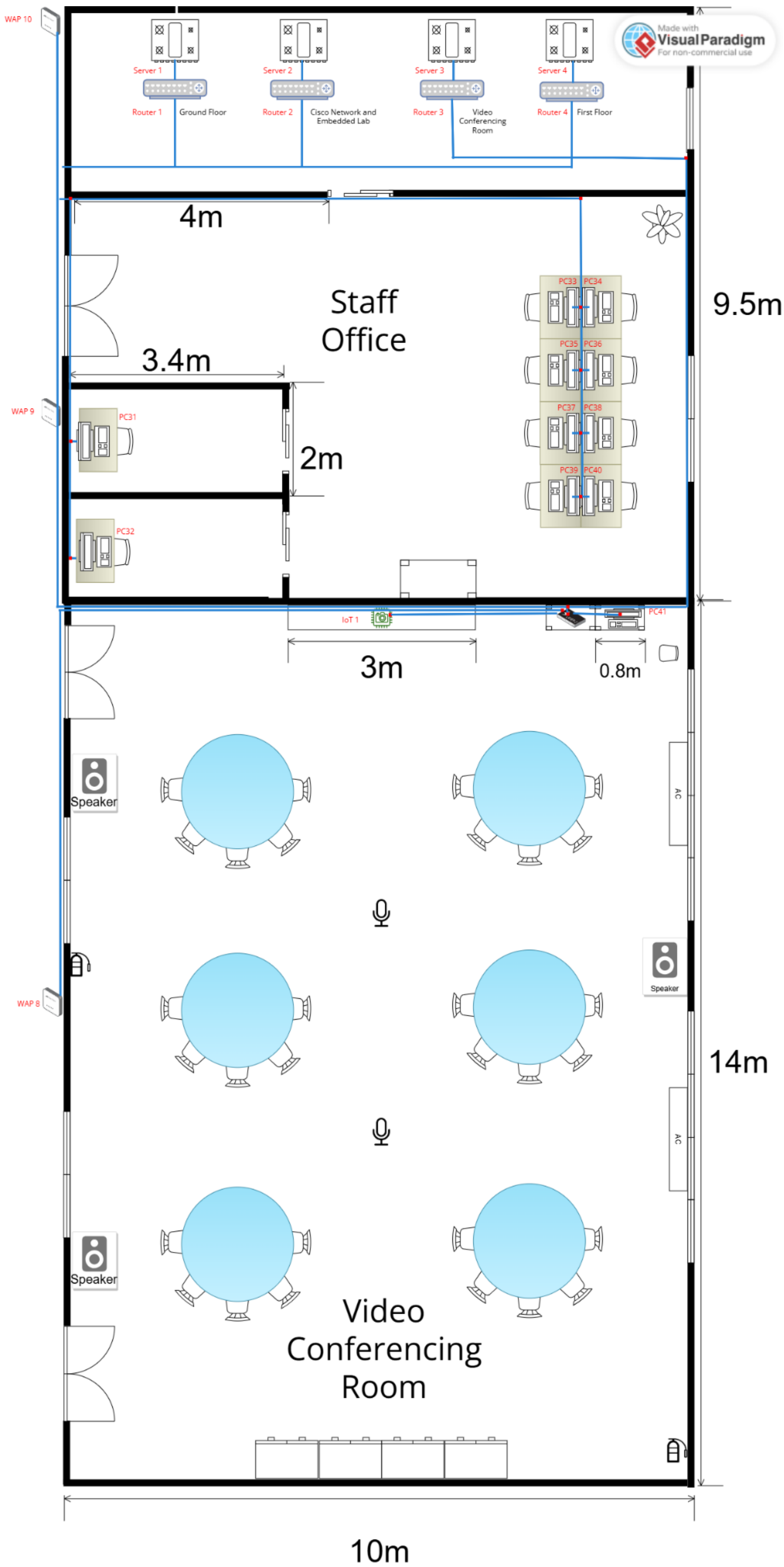

Figure 5.2.4 Server Room, Staff Office and Video Conferencing Room with label

Meeting Minutes #9

DATE/TIME	*22 Jan 2025 8pm*
LOCATION	Google Meeting
AGENDA	1. *Discuss about all the task from task 5* 2. Discuss about the network division 3. *Complete task 5 together.*
Meeting MC	Chau Ying Jia

ATTENDANCE		
NAME	**TIME**	**REASON FOR ABSENCE**
Poh Lok Yee	2000	-
Woo Cheng Shuan	2000	-
Chau Ying Jia	2000	-
Sabrina Heng Wei Qi	2000	-

MINUTES			
NO.	**ITEM DISCUSSED**	**IDEAS/SUGGESTIONS AND PERSON GIVING IT**	**PERSON IN CHARGE**
1	Discussion for task 5	*Chau Ying Jia brief us the information and rubrics in task 5* *All members read and understand requirements from task 5.*	Chau Ying Jia
2	Suggestion for network division	*All members suggest their prefer subnetwork division and decide the one best suits our design*	All members
3	IP Addressing Scheme Overview	*All members examine how IP addresses are currently assigned to each work area.*	All members
4	Task Distribution	*Ying Jia distributed the tasks for all members.*	All members
5	Proceed Tasks 5 together	*All members work on Words together to complete task 5.*	All members

6	Meeting Ended	*Meeting ended after all discussions have done* at 2300	All members